ETH zürich

Mobile Computing

Decision-Making

Stefan Mangold
stefan.mangold@inf.ethz.ch

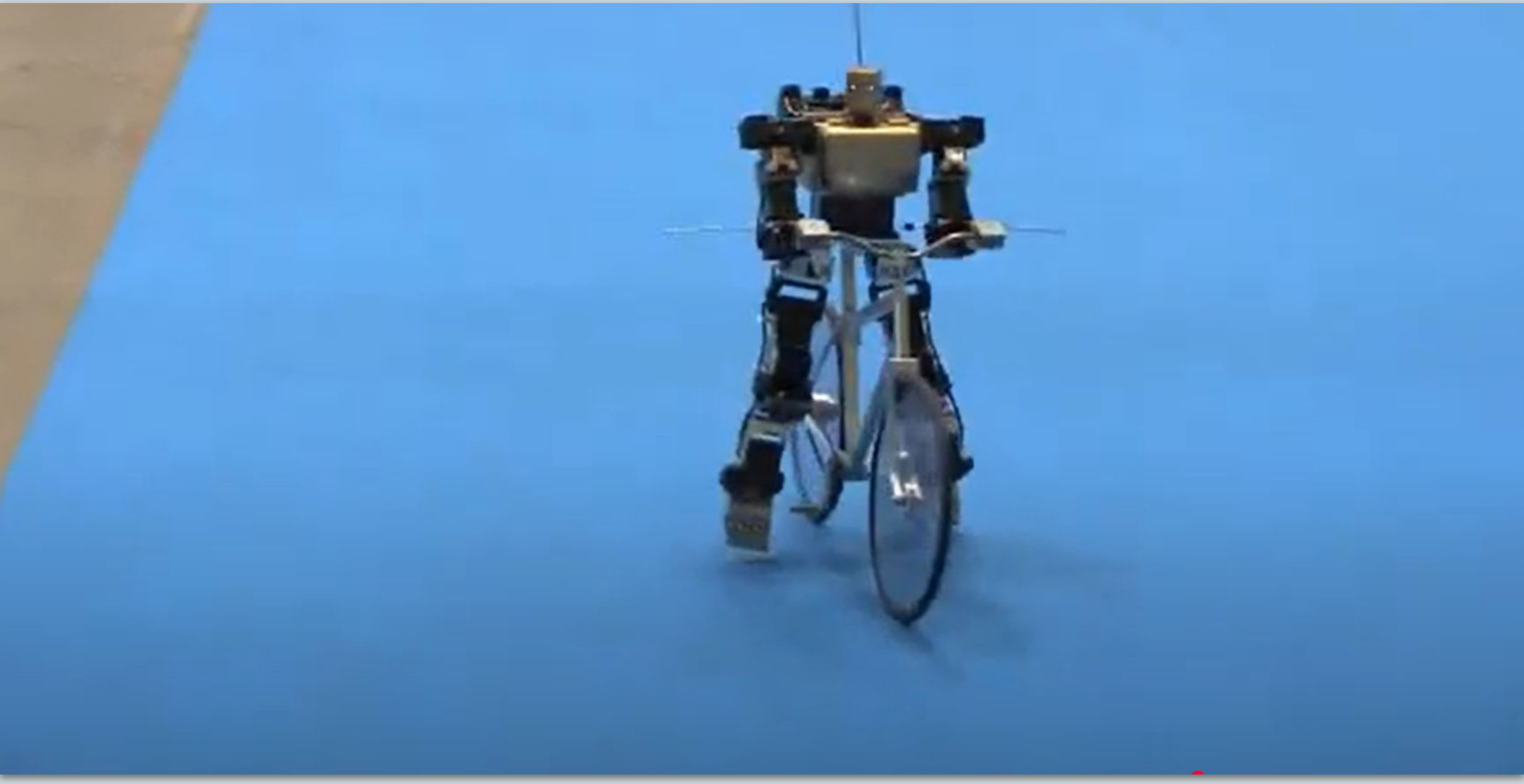
Course Outline

Introduction

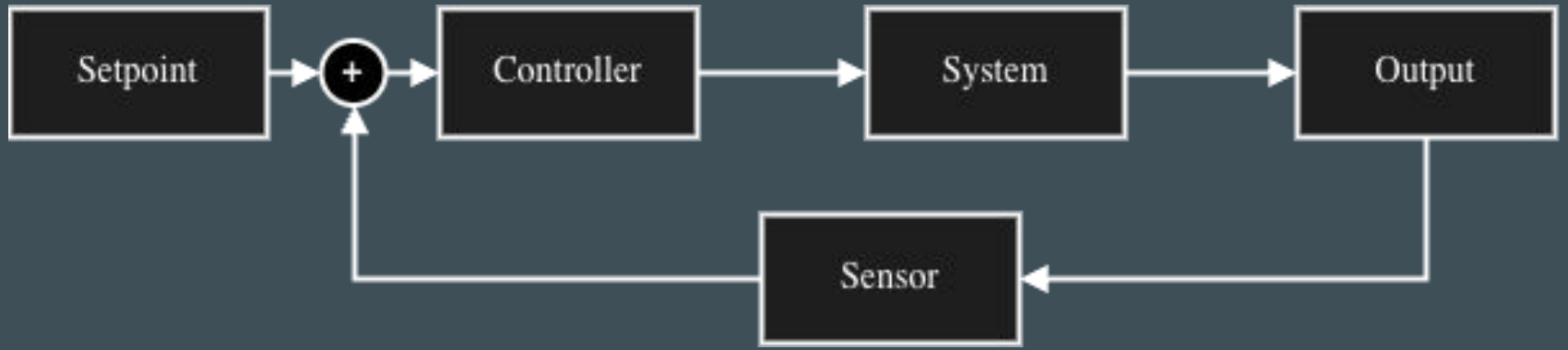
1. Data and Signals
2. Data Communication
3. Mobile Systems
4. Mobile Algorithms
5. Mobile AI
6. Cloud Computing

Tournament

Robot with Autonomous Controller



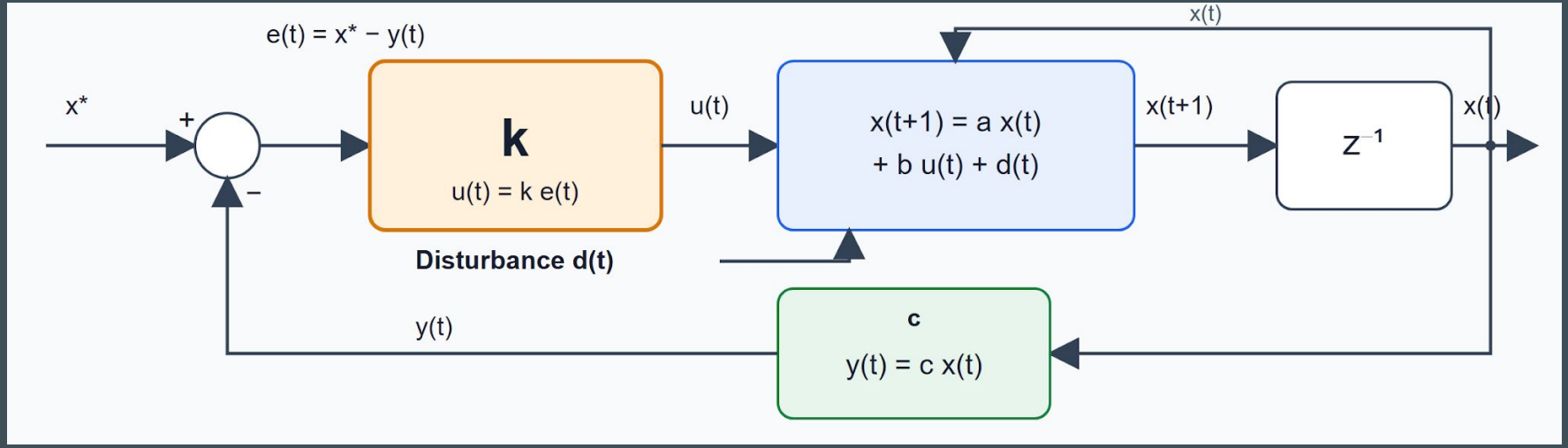
System Controller



Open-Loop Reflexes and Closed-Loop Control

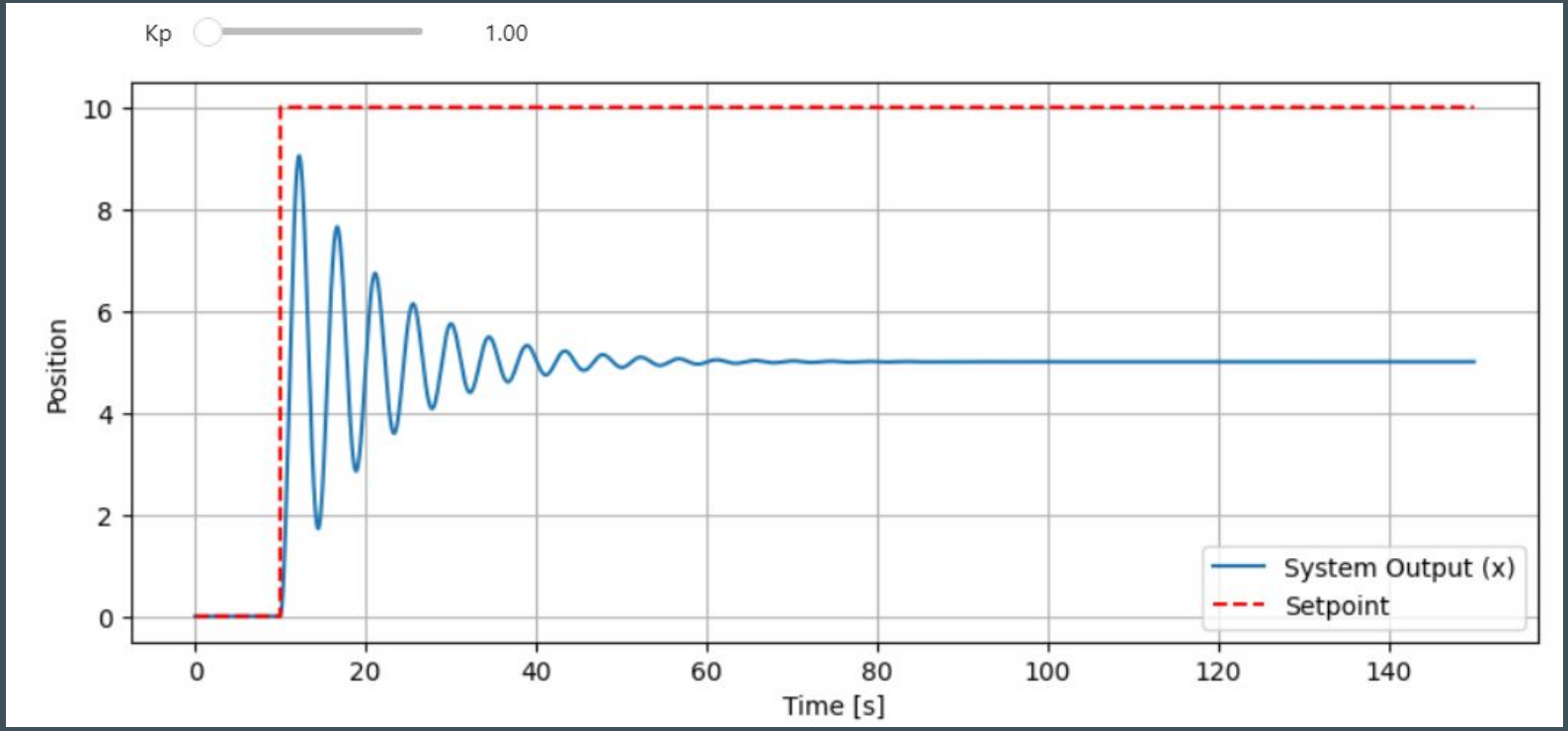
- Open-loop reflexes are pre-defined reactions
  - Direct mapping from input to output, no decision making, only reaction
  - Commands without taking feedback into account
- Closed-loop control processes measures and takes feedback into account
  - To bring system to a defined set-point and keep it there

Proportional Controller



Proportional Gain Kp

- The easiest and most common feedback control



Proportional Gain Kp

- For small gains, the system approaches the desired target, but it often does so slowly
- Increasing the gain speeds the response
  - Beyond that, system starts out faster, but it overshoots the target
  - In the end the system does not settle quicker than it would have with lower gain, but there is more overshoot
- Increasing the gain would eventually reach a point in which the system just oscillated around the target and never settled

Proportional Control: Model

- $x(t+1) = a\ x(t) + b\ u(t) + d(t)$
  - a: persistence; b: input effect; d: disturbance
- $y(t) = c\ x(t)$: sensor measurement
- $e(t) = x^* - y(t)$; $u(t) = k\ e(t)$

- k = Kp: current error only; no I or D term

Closed-Loop Dynamics

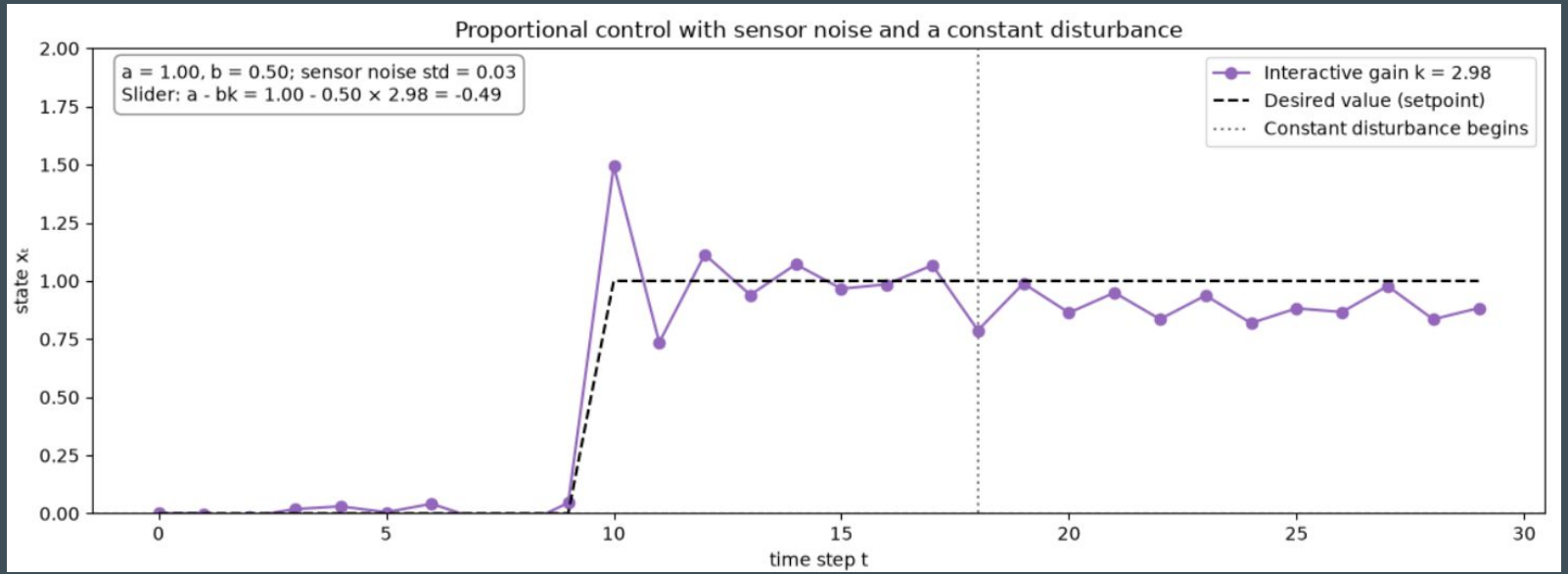
$$x(t+1) = a\,x(t) + bk[x^* - c\,x(t)] + d(t)$$

$$x(t+1) = (a - bkc)x(t) + bkx^* + d(t)$$

- Therefore:

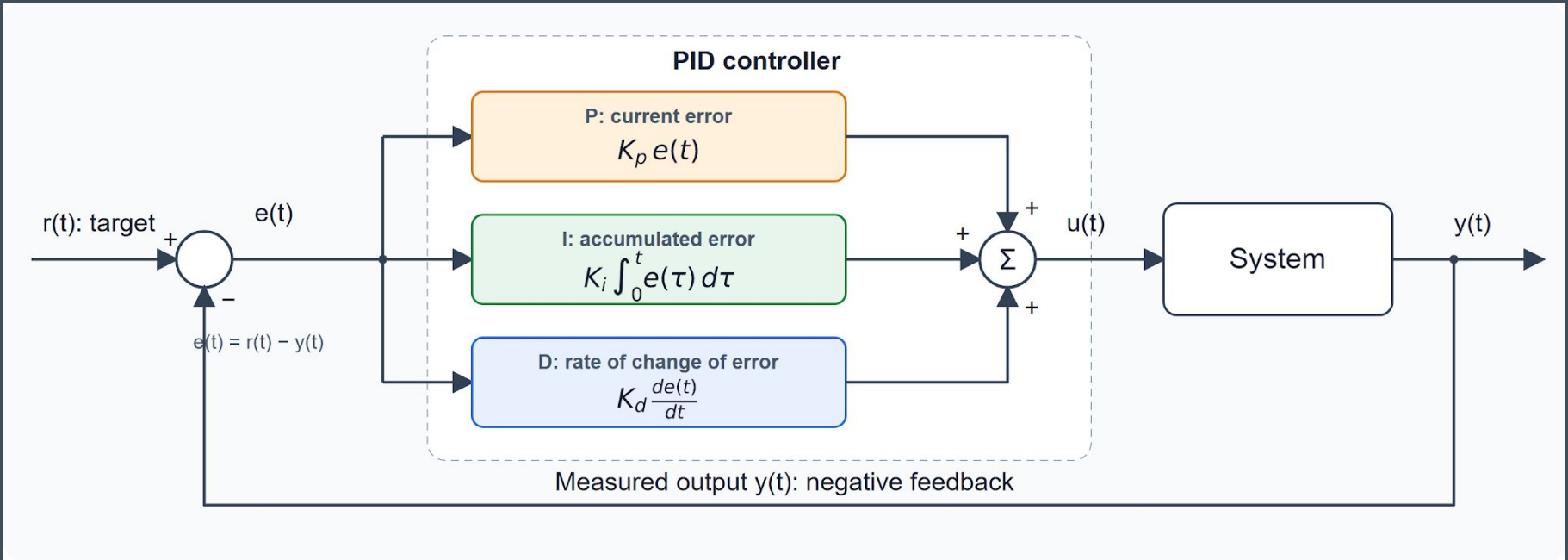
a – bkc governs decay or growth

Stability is an important design principle



- Typical questions
  - How responsive is the controller to changes in the set-point?
  - Does it settle down or does it oscillate?

Proportional, Integral, Derivative



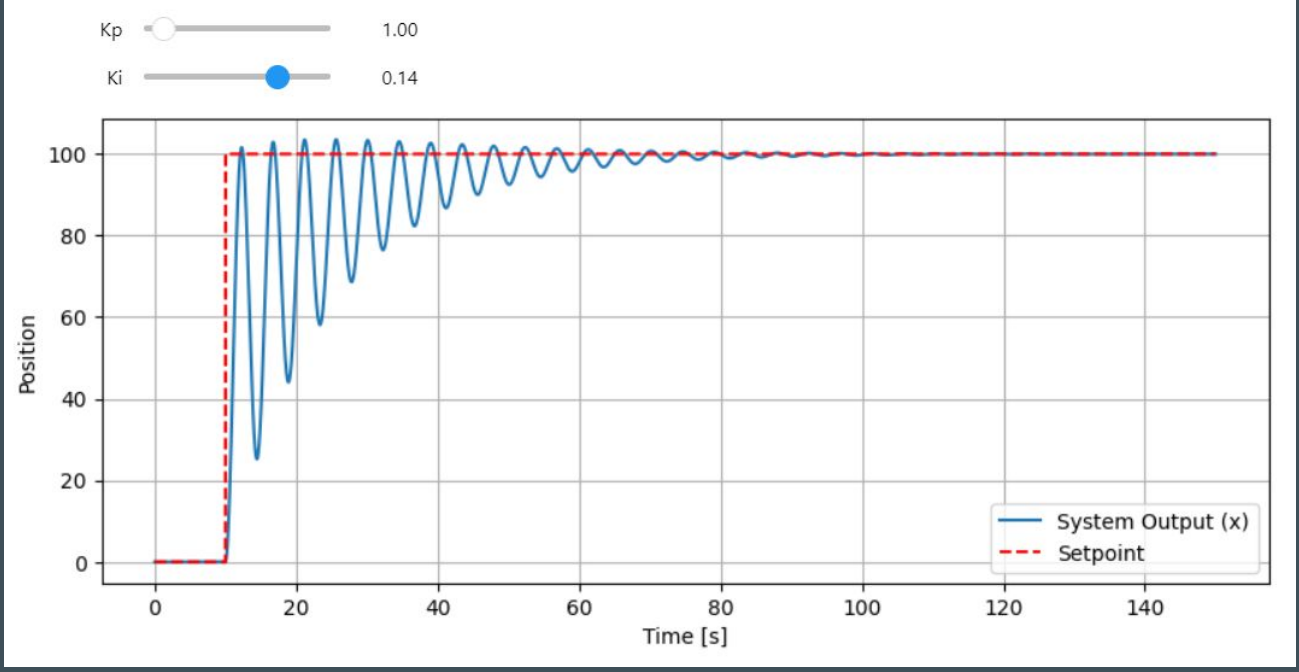
Intuitive Understanding

```
out = Kp*error + Ki*integral + Kd*derivative
```

- Kp determines how fast the error is tracked
  - The system can lose stability with too large gains
- Ki can be used to reduce the proportional offset
  - Might introduce error accumulations - high frequencies attenuated
- Kd enables fast reaction on error changes
  - Amplifies noise, but can be addressed with low pass filtering
  - Low frequencies (slow changes) are attenuated

Integral Gain Ki

- Integral control is used to add long-term precision to a control loop

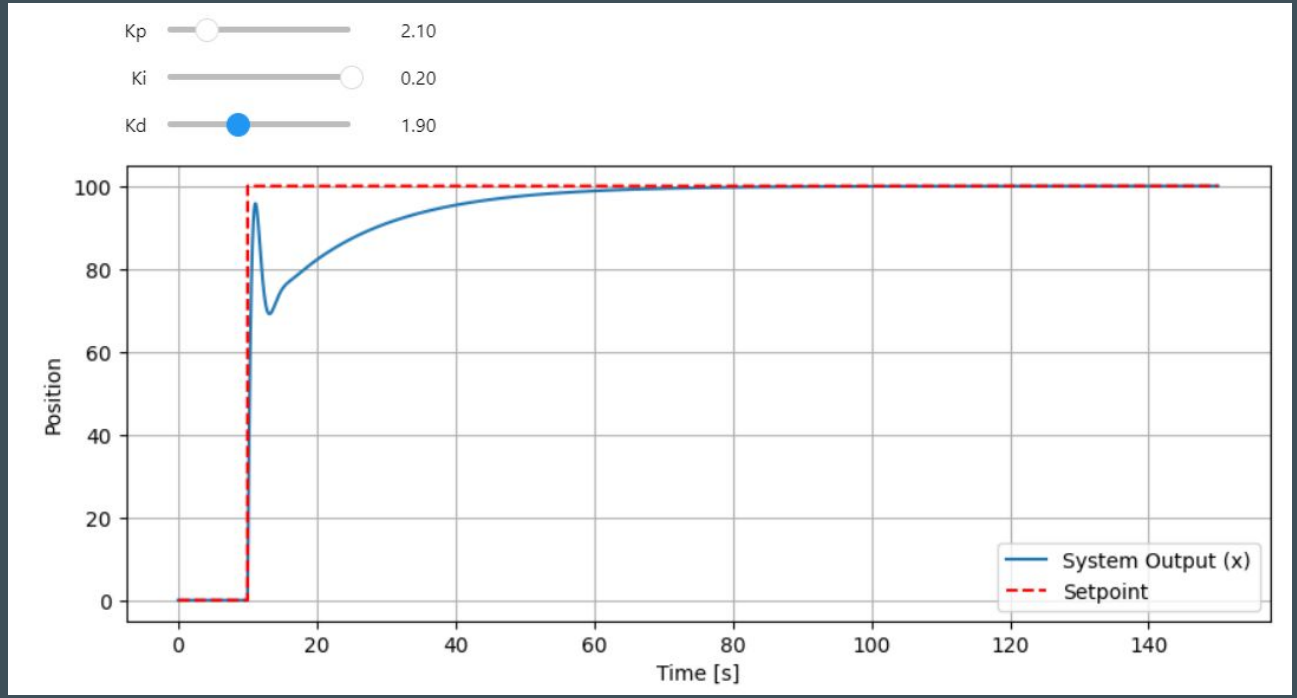


Integral Gain Ki

- Proportional control addresses present system behavior
- Integral control addresses past system behavior
  - Integrator accumulates and retains historical state data
  - Eliminates steady-state and short-term output errors
- Accumulated memory can introduce system instability
  - Controller response experiences time lag
- Instantaneous proportional feedback is required to maintain stability

Derivative Gain Kd

- Derivative control reduces oscillations



Derivative Gain Kd

- Some element that predicts the future behavior is required to help stabilizing a system

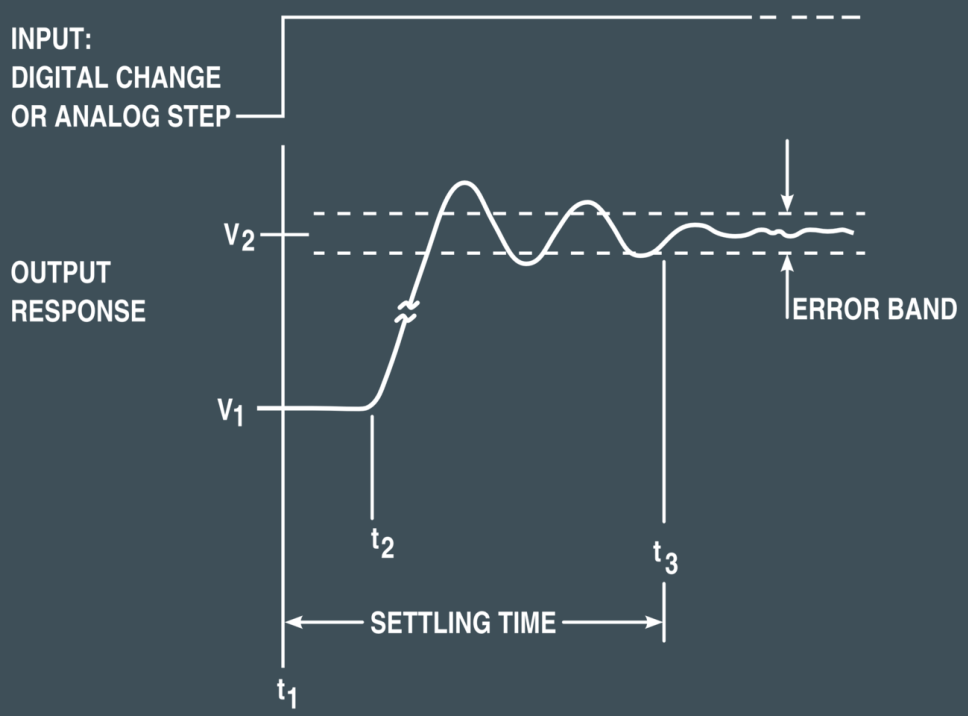
  → Differentiator
- The differential term is the last input value minus the current value
- This predicts where the system will be next

PID Pseudo Code

```
01  integral = 0
02  previous_error = target - read_sensor()
03
04  while True:
05      measured_value = read_sensor()
06      error = target - measured_value
07      integral = integral + error * dt
08      derivative = (error - previous_error) / dt
09      out = Kp*error + Ki*integral + Kd*derivative
10      apply_control(out)
11      previous_error = error
12      wait(dt)
```

Design Parameter: System Settling Time

- The time required for an output to reach and remain within a given error band following some input stimulus



Design Parameter: Sampling Rate (Rate = 1 / dt)

- Low control performance if sampling rate is too low, because of the added delay of the sampling
- Too high sampling rate might create problems
  - Noise in the differentiator, overflow of the integrator
- Rule of thumb for control systems:

Set sample time dt =
1/10th ... 1/100th of system settling time

Noise

- Central Limit Theorem - many independent sources of noise
  - Spread across the frequency spectrum: AWGN
  - Control commands are at lower frequencies
- Proportional control passes noise energy through
- Integral control averages the noise input signal over time, which tends to reduce the energy
- Differential control might amplify higher frequency components and the noise energy

Regulatory Acceptance of PID

- Deterministic behavior
  - PID controllers always produce the same output for the same input - no randomness, learning, or hidden states
- Predictable and analyzable
  - Their behavior can be fully described with mathematical models and proven stable with classical tools
- Transparent design
  - Only three parameters (P, I, D), each with a clear physical meaning and easily explained tuning
- No black box
  - Every design choice and output can be justified and verified

Regulatory Acceptance of PID

- Known failure modes
  - Poor tuning may cause overshoot or oscillation, but not chaotic or unpredictable behavior
- Regulatory acceptance
  - Agencies prefer deterministic, verifiable systems for medical and safety-critical devices
- Trusted in critical applications
  - Used in insulin pumps, artificial pancreas systems, ventilators, infusion pumps, and industrial control

Exercise: PID Controller

- Open in your course folder: 02_Notebooks/04_Mobile_Algorithms/ 04.2_PID_Controller.ipynb
  - Run the cells and answer the questions.
  - Save the completed notebook.


SPEED 1863 KM/H
ALTITUDE 5 KM
LOX
CH4
T+00:06:20
STARSHIP FLIGHT TEST
SPEED 11554 KM/H
ALTITUDE 130 KM
LOX
CH4
0:03 / 1:14

Exercise: Rocket Landing with PID

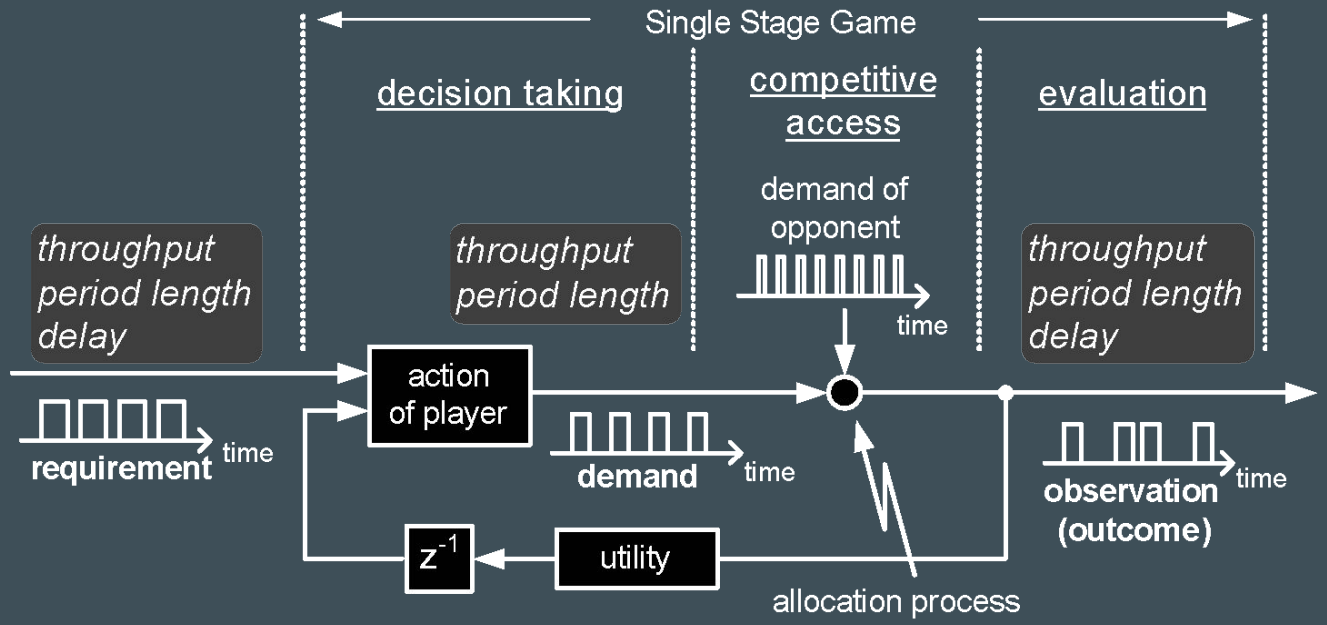


Exercise: Rocket Landing with PID

- Open in your course folder: 02_Notebooks/04_Mobile_Algorithms/ 04.4_Rocket_Landing_With_PID.ipynb
  - Run All; tune both PID controllers in the cockpit.
  - Challenge: highest wind with about 4 successes in 5 attempts.
  - Complete Section 4.6, save, and upload to Moodle

Spectrum Sharing Games

- Repetitive competitive access to shared spectrum
  - Stochastic evolutionary game dynamics in finite populations
  - Observed QoS as outcomes of a single stage game

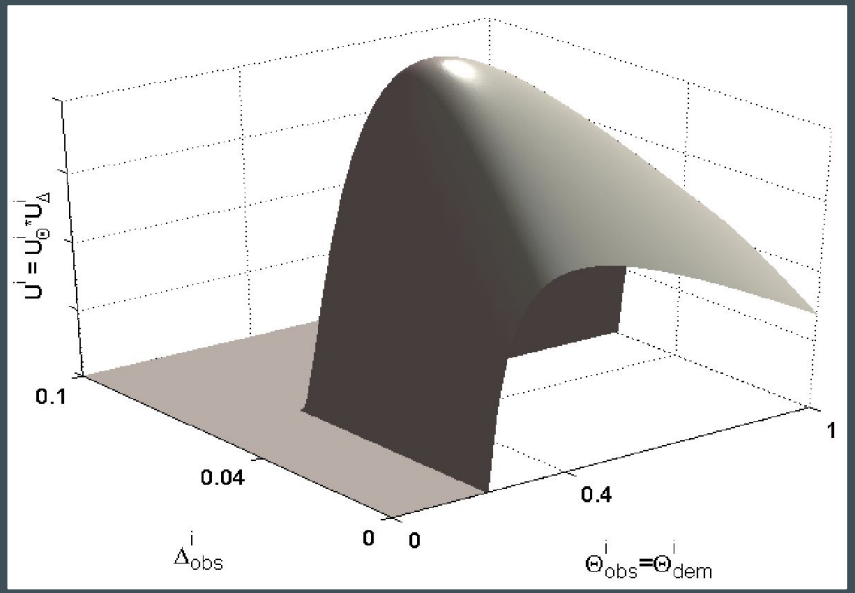


Game Theory in Mobile Computing

- Game concepts can help to analyze a competition scenario
  - How does natural selection lead to cooperative behavior?
  - Rational behavior and predictability
  - Self interest (payoff and utility function)
  - Nash Equilibrium and Pareto Efficiency
- A cooperator pays a cost for another individual to receive a benefit
- In network systems, it is not necessarily the most desirable approach to try to achieve fairness

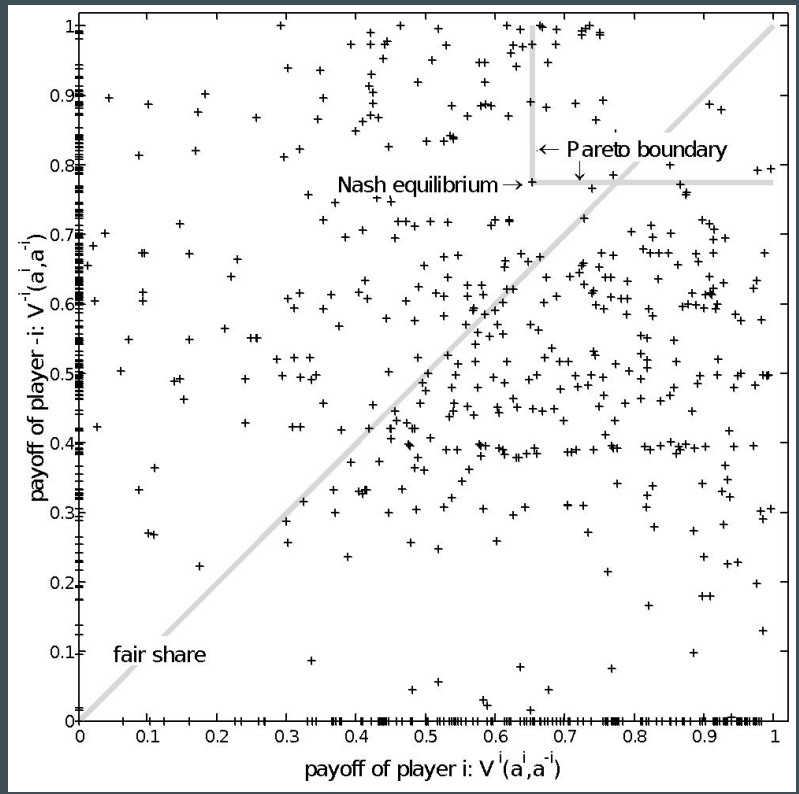
Utility Function

- Player (index “ i ”) attempts to meet its requirement
  - Therefore, players attempt to maximize the observed payoff (outcome)
  - The current payoff is high when the observation meets (or exceeds) the utility



Pareto Efficiency and Bargaining Domain

- Rational decisions might lead an repeated interaction to equilibrium
- Equilibrium: An action profile that is not changed by one player alone, once achieved
  - Action profiles with higher payoffs may still exist
  - If not, then the NE is Pareto efficient



Interaction as a Single Stage Game

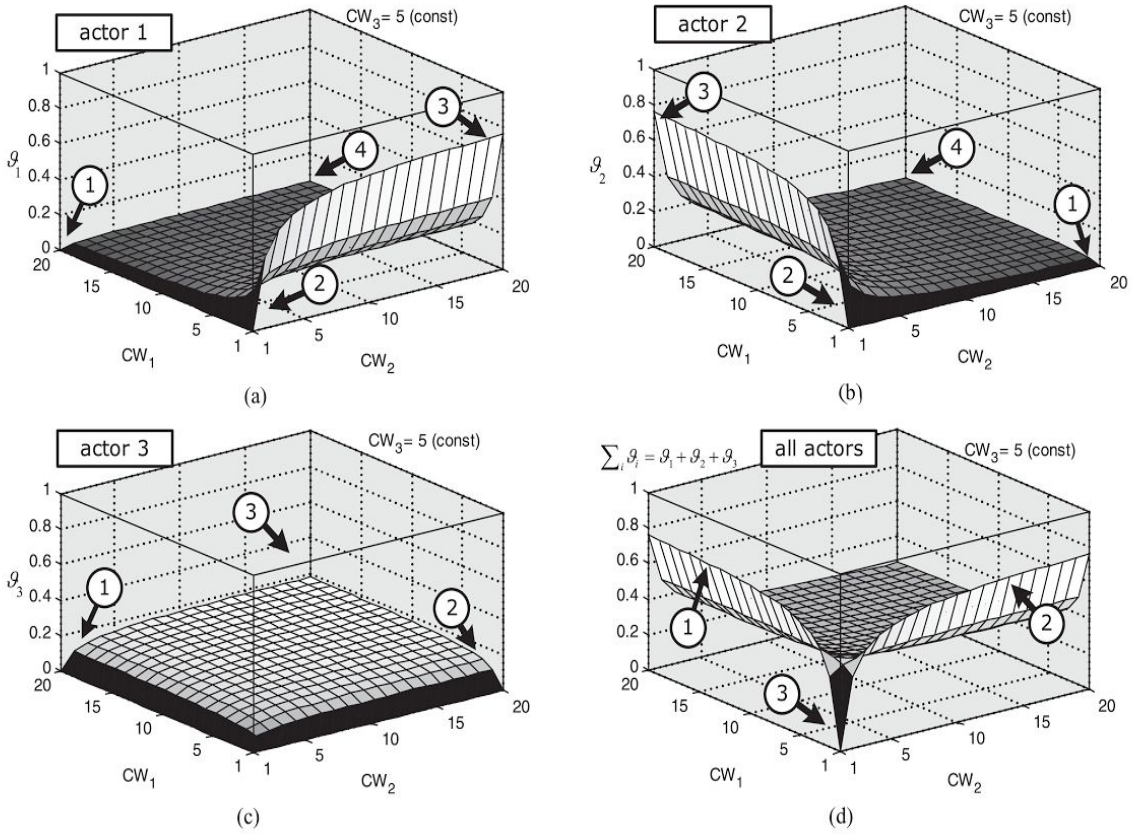
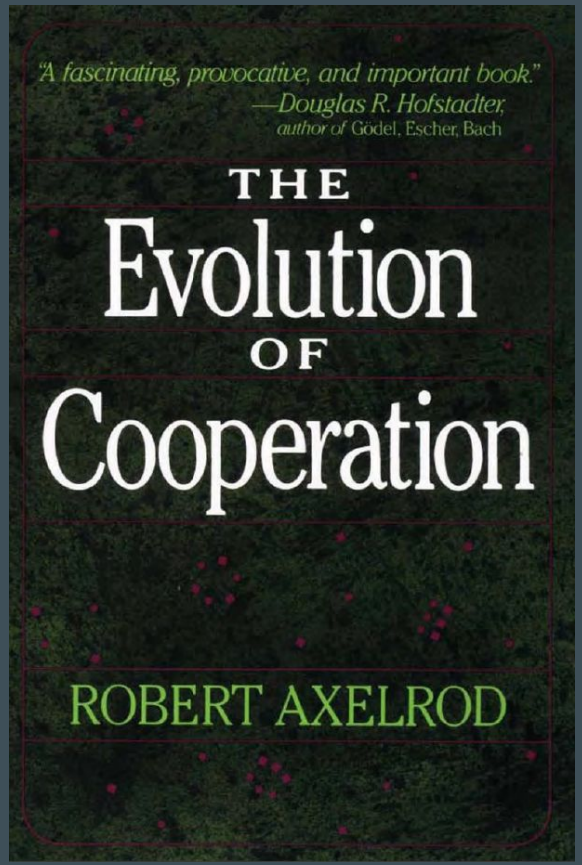


Figure 2. Observed shares of capacity ϑ_i and $\Sigma\vartheta_i$ for the three actors, with varying contention window sizes for actor 1 and actor 2. (a) observed share of actor 1; (b) observed share of actor 1; (c) observed share of actor 3 with $CW_3 = \text{const} = 5$; (d) sum of all actor's shares.

Multi-Stage Games and The Evolution of Cooperation

- Repeated interaction of rational players might end in long-term cooperation.
- Even with self-interest of players
- A “non-cooperative” game is a game where cooperation is achieved through self-interest, not through bargaining
- Cooperation might not lead to the most efficient outcome, but often to a balanced result

The Evolution of Cooperation



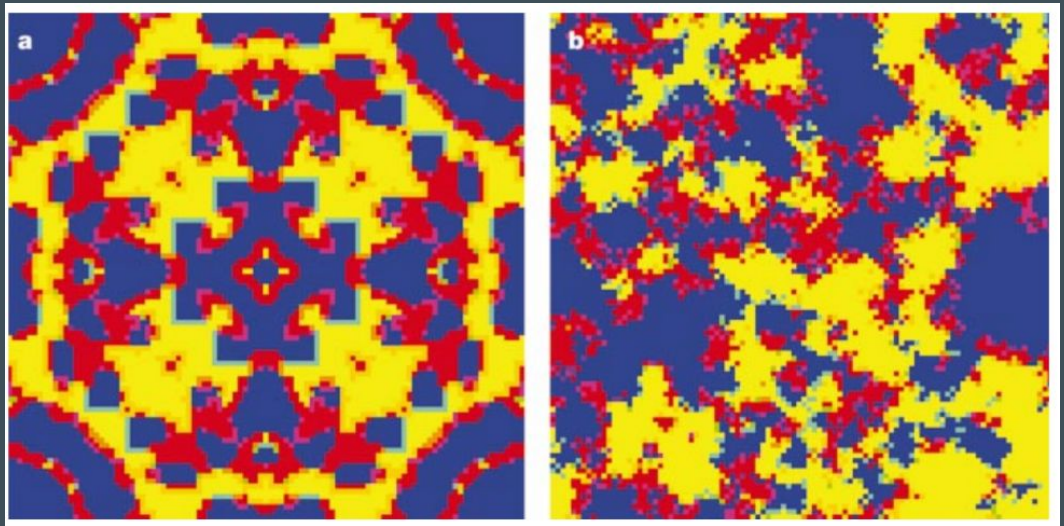


Figure 1 **Evolutionary kaleidoscope.** a, **Players in a public-goods game are represented as cells distributed spatially on a grid. Cooperators are shown in blue, defectors in red and loners in yellow. Players may change their strategy in each round of the game. For this simulation, all strategy updates are synchronous. Intermediate colours indicate players who have updated their strategies in the last round of the game.** b, **Another simulation shows the effects of asynchronous strategy updating and small amounts of noise in the system. In both cases, there is coexistence between cooperators, defectors and loners. Szabó and Hauert[5] show that the dynamics of such systems have much in common with models in statistical physics.**

NATURE | VOL 419 | 17 OCTOBER 2002 | www.nature.com/nature

Types of Social Actions

- Weber’s basic classification defines four types of motivation for social action
- From Max Weber: Wirtschaft Und Gesellschaft. 2, Verm. Aufl. ed. Vol. Abt. 3, Ed. 2. Tübingen: Mohr, 1925. Print. Grundriss Der Sozialökonomik
  - Affective social action (based on emotional state)
  - Traditional social action (guided by custom)
  - Technocratic rational social action ("zweckrational")
  - Value-oriented rational social action ("wertrational")
- Value-oriented actors limit need for regulation

Example: Tit for Tat

- Scenario: The repeated Prisoner's Dilemma:
  - Multi-stage behavior in which a player mimics the action of the opponent, after cooperating in the first stage
  - Value-oriented / cooperative social action: does not attack first
  - Technocratic social action: forgives earlier attacks
  - Affective social action: attacks when attacked
- Tit-for-tat does whatever the opponent has done in the previous round [Source]
  - Cooperate if the opponent has cooperated
  - Defect if the opponent has defected

Mobile A.I.

- With the rise of LLMs, we may finally be able to tackle the artificial complexity that has long caused so many challenges
- LLMs may finally help us cut through to a true simplification of our systems

This is the idea behind “Mobile A.I.” !

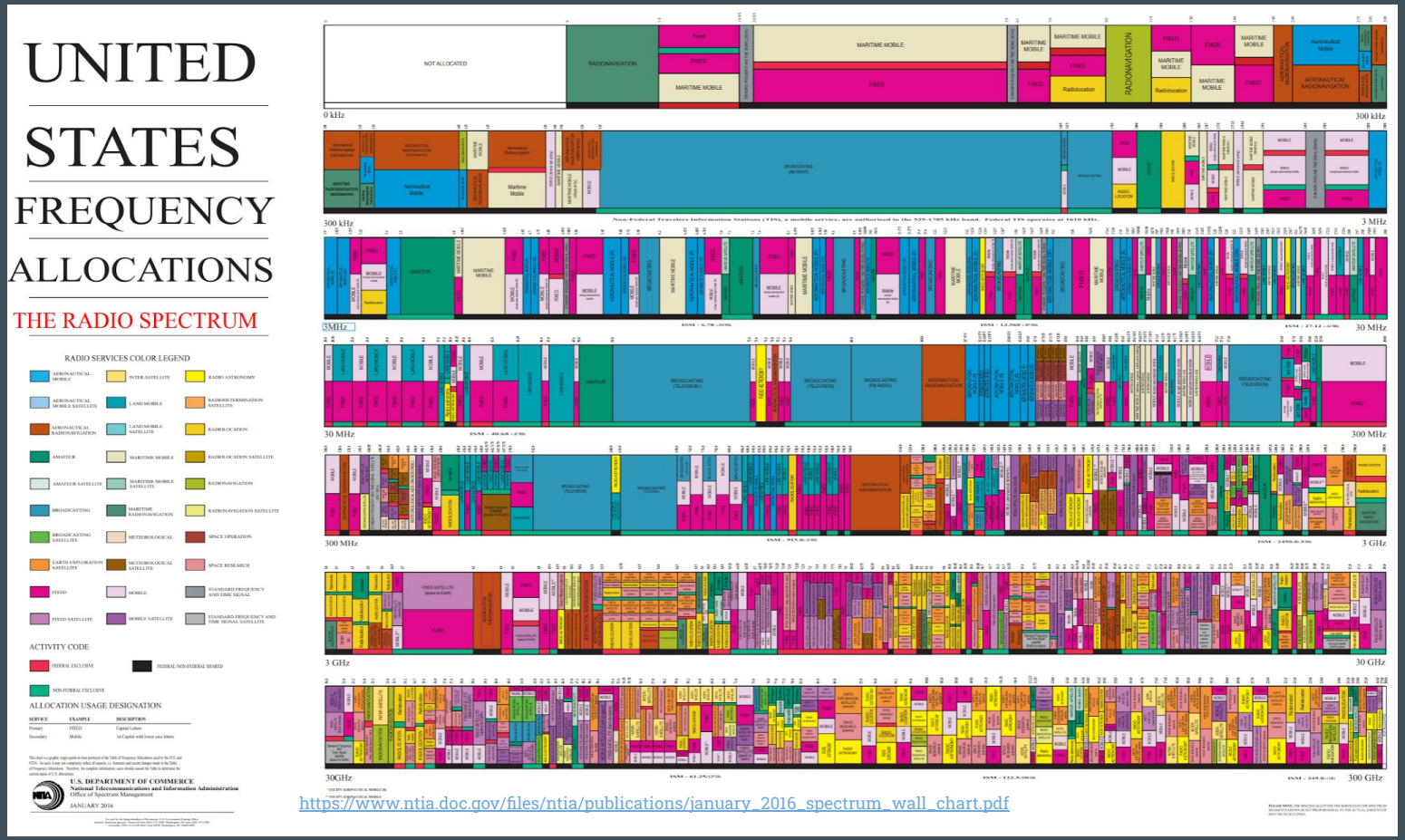
Exercise: Tit-for-Tat Spectrum Management

- Open in your course folder:
  02_Notebooks/04_Mobile_Algorithms/
  04.3_Tit-for-Tat_Spectrum_Management.ipynb
- Run the cells and answer the questions.
- Save the completed notebook.

Radio Spectrum Etiquette

- Spectrum occupancy and measurements
- Rules for spectrum sharing
- DARPA XG and reasoning engines

FCC Radio Spectrum



https://www.ntia.doc.gov/files/ntia/publications/january_2016_spectrum_wall_chart.pdf

Radio Spectrum Measurement

- We conducted measurement of how overloaded the spectrum really is
  - Use a simple omni-antenna and a spectrum scanner
  - Measure in all directions, for a long time





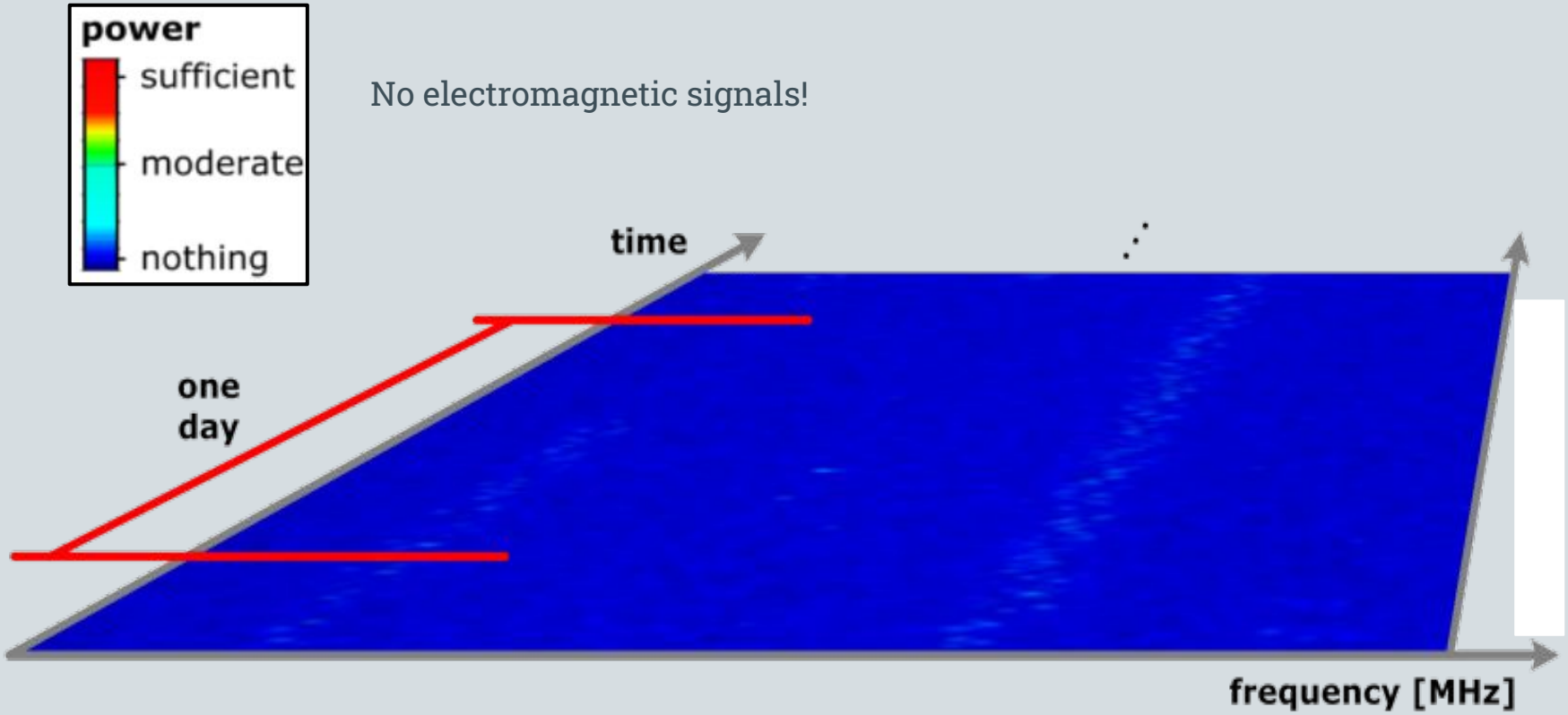


Measurements

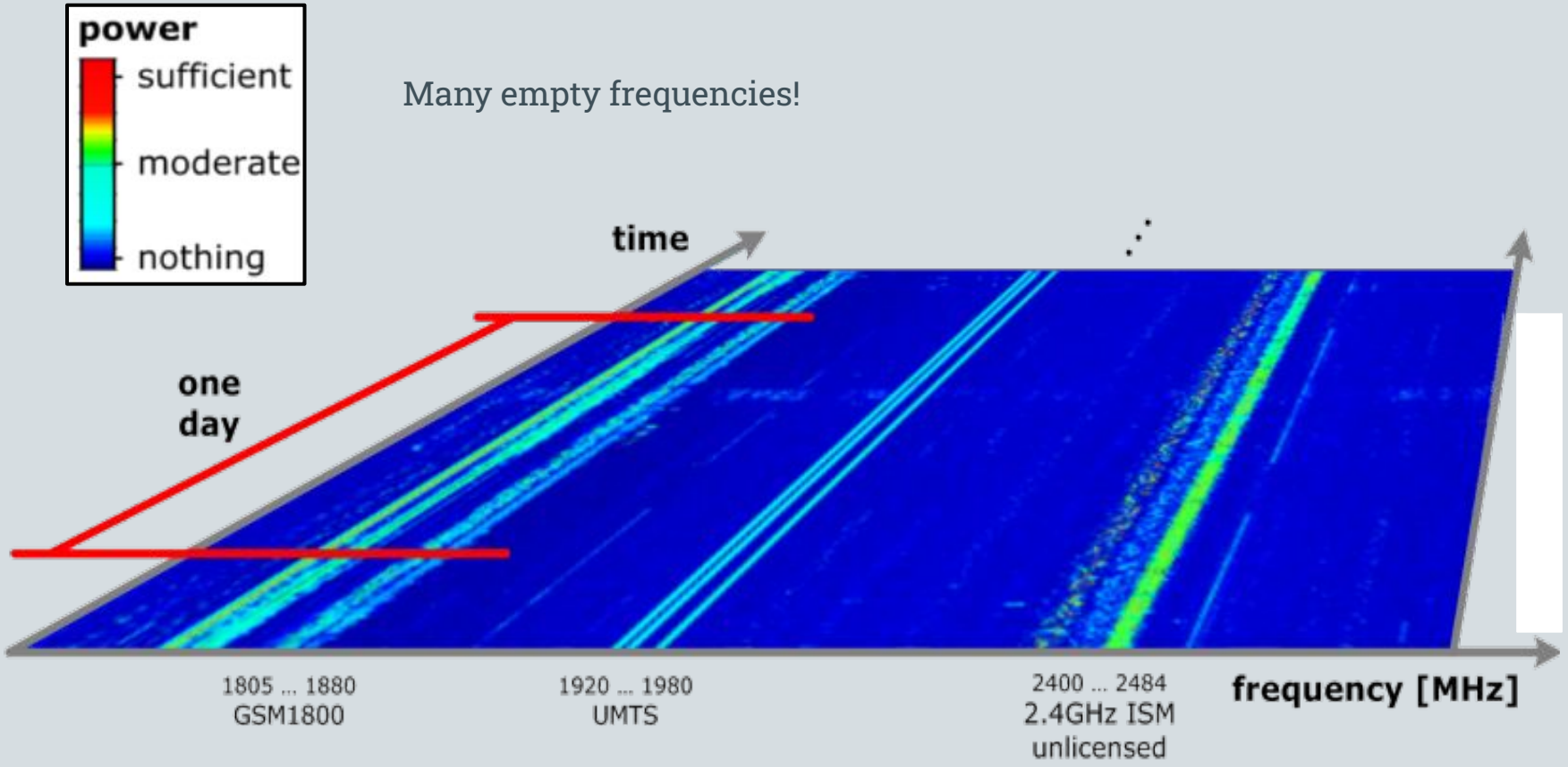


in office & basement shelter

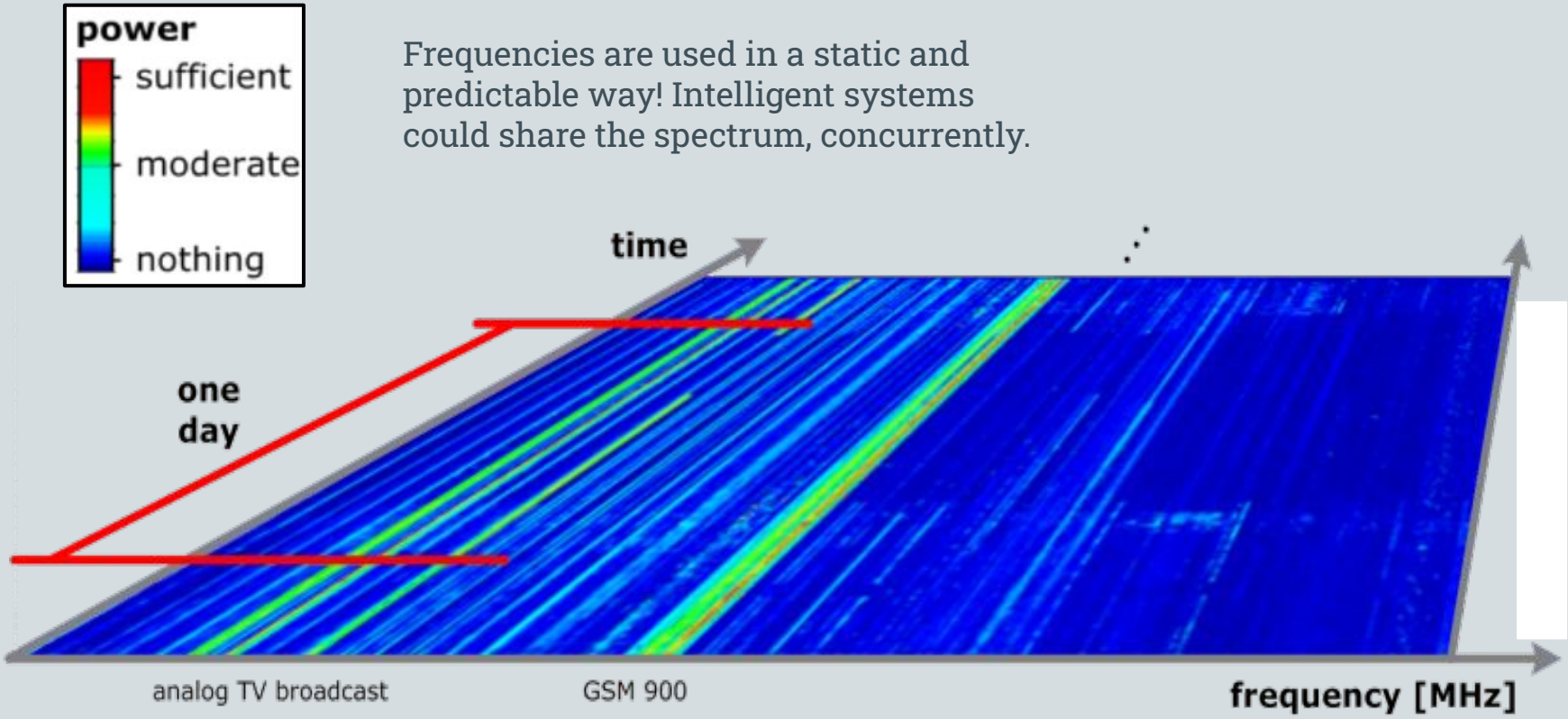
In a Shelter in Berne, Switzerland



Office Space 1.5 GHz ... 3 GHz



Office Space, 0.5 GHz ... 1.5 GHz

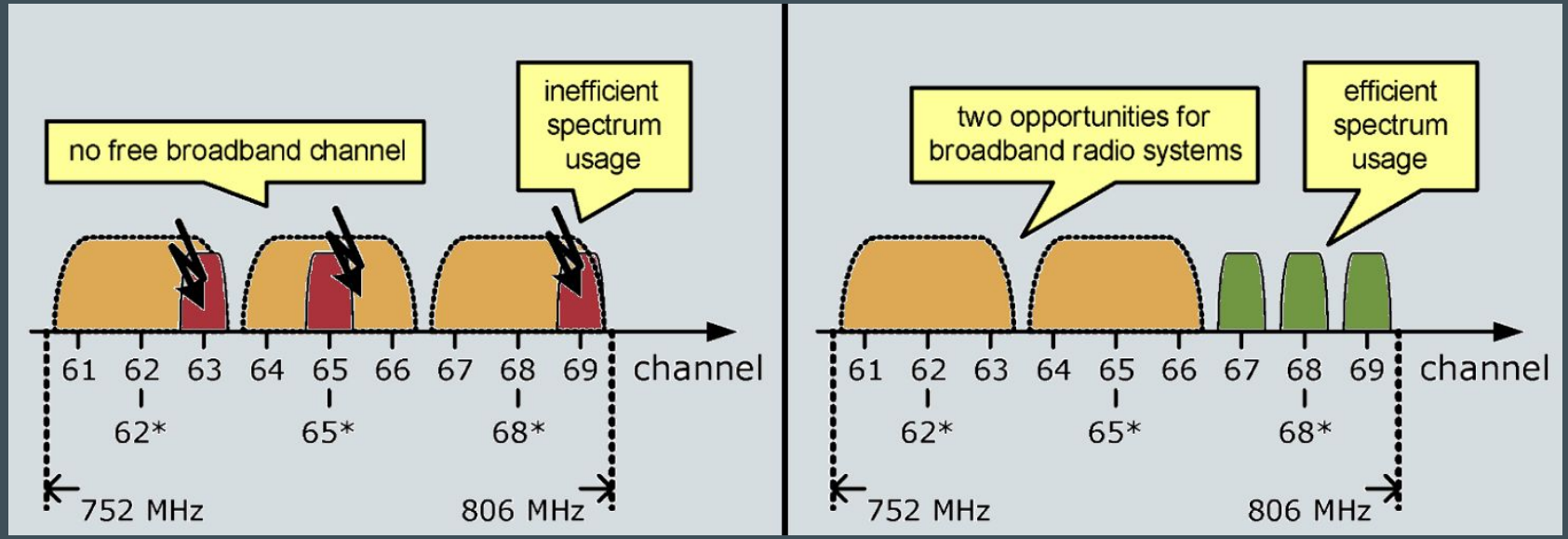


Spectrum Etiquette

- Smart devices with social awareness
- Primary aim is to promote consideration for others
- Voluntary rules for socially acceptable behaviors
- Location-, device-, and scenario-dependent guide lines
  - Framework for adapting the communication parameters
  - … such as, protocol, radio signal, and antenna characteristics
- Several techniques can be applied to achieve this
  - Logics, reasoning, inference

Example

- Two radio systems with different bandwidths
- Left: random selection
  Right: rule-based etiquette



DARPA NeXt Generation (XG) Program

- XG was a (not entirely successful) attempt for a technology development project
  - sponsored by DARPA's Strategic Technology Office
- Objective:
  - “To develop the technologies and concepts to dynamically redistribute allocated spectrum along with novel waveforms ...”
  - “... to provide improvements in assured military communications in support of a full range of worldwide deployments"
- XG might have been too ambitious (at its time, before 2010)

DARPA XG Reasoning Engine

- A mechanism to create previously unknown assertions, and to demonstrate their truth
  - The reasoner creates an assertion upon stimulus
  - Starts with premises (the existing knowledge, spectrum regulation)
  - Uses valid inference rules (the arguments)
  - Infers a conclusion (the actual action)
- Reasoning engines
  - Web Ontology Language OWL, first order predicate logic
- BENEFITS: Traceable decision making and patent infringement detection

